Step-Up and Step-Down Rheology of Human Blood: Data, Methods and Findings

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Written report accompanying the presentation *Step_UpDown_Findings*

Summary. We analysed 136 step tests on 11 human blood samples, in which a rheometer suddenly changes the shear rate and holds it while the stress response is recorded. After verifying every test against its protocol label and removing copied blocks, each response was reduced to a small set of interpretable transient features, and a published thixo-elasto-visco-plastic model was fitted to every sample. The model reproduces the published time scale and stiffness of blood (median structural time 0.64 s; median elastic modulus 0.27 Pa) but misses a slow 10–60 s recovery after step-downs. Transient and stable phases differ in all 11 samples; step size is encoded by different features for each direction; and in all 22 matched up/down pairs, the step-down has the larger undershoot. A pre-specified direction rule failed (55 % accuracy). We recommend undershoot, initial slope and settling time as cheap candidate features for real-time use. With 11 independent samples, all results are exploratory.

1. Introduction

In a step test, a rheometer holds blood at one shear rate (how fast neighbouring layers of fluid slide past each other, in s^{-1}) and then suddenly switches to a different rate and holds it. The shear stress, the force per unit area with which the blood resists the flow (in pascals, Pa), is recorded for about a minute after the switch. Blood does not respond instantly: red blood cells form stacked aggregates called rouleaux, which break up when flow speeds up and rebuild when it slows down, so the stress evolves over time (Pincot et al., 2023).

> **Worked example.** in one test the rate is held at 5 s^{-1} and then dropped to 0.1 s^{-1}. Before the switch the stress is about 0.046 Pa; a minute later it has settled at about 0.0056 Pa. The path between these two values is what this report studies.

We address four questions:

- Can simple features of the response tell how large the step was (step magnitude)?
- Can they tell whether the rate went up or down (step direction)?
- Can the transient part of a test be distinguished from its stable part?
- Which small set of features would be worth computing in real time?

The work follows the project's data science workflow: ingestion, verification and cleaning, exploratory analysis, feature extraction, modelling and validation. Analysis choices were fixed before results were inspected, and all statistics treat the blood sample, not the individual test, as the independent unit.

2. Data

2.1 Sources and preparation

The data are ARES-G2 strain-controlled rheometer measurements on human blood (double-wall Couette geometry, 37 °C, pre-shear at 300 s^{-1} for 30 s before each test; Pincot & Armstrong, 2021), published by Armstrong and Horner on Mendeley Data and supplied as Excel workbooks. The counts were reached as follows:

- 15 workbooks were received.
- 4 were set aside: two copies of a summary-only file with no raw time series, and two copies of an aspirin study containing only steady-flow data. This leaves 11 blood samples.
- Each workbook stores several tests side by side; these were split into separate step tests using the repeated *Step time / Stress / Shear rate* column headers.
- 7 blocks had a stress trace identical to another block and were removed as copies, leaving **136 step tests**.

  Worked example. in the 14DEC workbook, the blocks labelled 0.1 → 10 and 0.1 → 20 contain exactly the same stress values as the 0.1 → 5 block, so only the 0.1 → 5 block is kept. In DEC17 at 24 h, the 0.1 → 1 block repeats the stress of the 0.1 → 0.5 block although its shear-rate column differs: only the stress column had been copied.

Most tests contain 65 points spaced logarithmically in time over 63 s; six high-resolution recordings (all 5 → 0.25 s^{-1}) contain about 4,200 evenly spaced points. For comparability, the high-resolution recordings were averaged into the same logarithmic time bins as the others before features were computed.

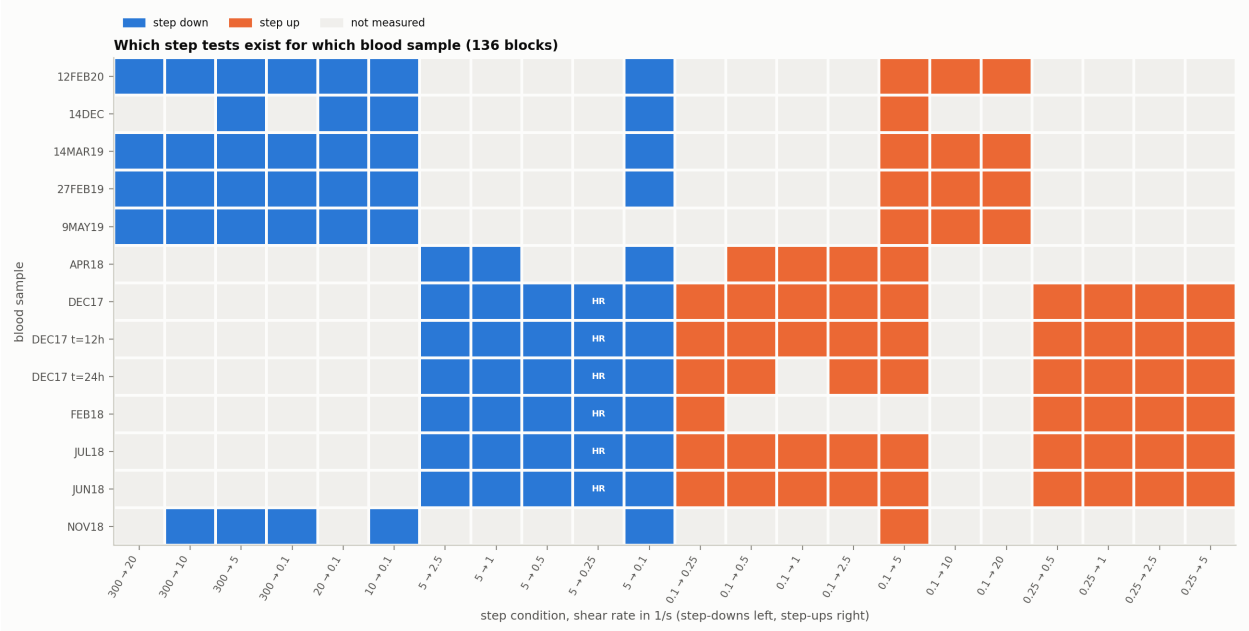


Figure 1. Data map. Vertical axis: blood samples, one row per workbook (DEC17 at 0, 12 and 24 h of storage). Horizontal axis: step conditions written as starting rate → final rate (s^{-1}); step-downs left, step-ups right. Blue: step-down measured; orange: step-up measured; grey: not measured. HR: high-resolution recording.

Figure 1 reveals a structural feature of the data. The top five samples were tested mostly with large step-downs from 300, 20 or 10 s^{-1}, while DEC17, FEB18, JUL18, JUN18 and APR18 were tested with steps between 5 and 0.1 or 0.25 s^{-1}, the protocol used in the published papers. Each sample was tested with only one of these two protocol families, so step size is partly tied to the sample. Every comparison of step size is therefore made within a *family*: the same sample and the same starting rate. Twenty-two *matched*

pairs, in which the same blood was stepped from a to b and from b to a (for example 5 → 0.1 and 0.1 → 5), provide a fair basis for comparing directions.

2.2 Verification against the protocol

Before any feature was computed, each test was checked against its label. The applied starting rate was taken as the median of the first three points (recorded before the motor moves) and the final rate as the median of all points after 1 s.

Formula. relative error = | measured rate ÷ labelled rate − 1 |. A test passes if this is at most 0.05 (5 %) for both the starting and the final rate.

Worked example. 12FEB20, labelled 0.1 → 20, has a measured starting rate of 0.1051 s^{-1}: 0.1051 ÷ 0.1 − 1 = 0.051, just above 5 %, so it fails. 14MAR19, labelled 0.1 → 10, starts at −0.30 s^{-1}, so the sample was not at its stated starting state. APR18, labelled 5 → 0.1, ends at 0.50 s^{-1}: 0.50 ÷ 0.1 − 1 = 4, a 400 % error, so it is really a 5 → 0.5 step.

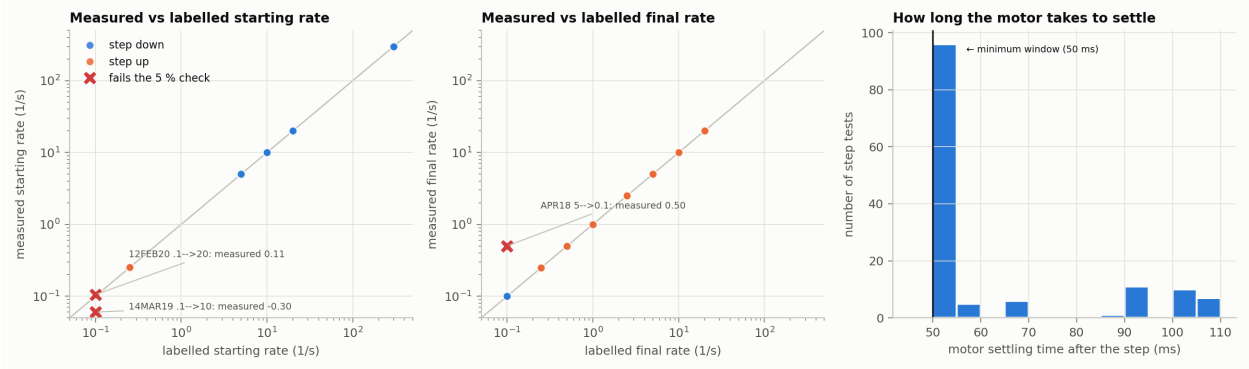


Figure 2. Step verification. Left and middle: labelled starting and final rate (horizontal, s^{-1}, log scale) against the rate actually applied (vertical, same scale); points on the diagonal match; red crosses fail the 5 % check. Right: time for the motor to settle at the new rate (horizontal, ms) against the number of tests (vertical).

In total 133 of 136 tests pass (Figure 2); the three failures are excluded from all later analysis. The motor reaches the new rate within 50–55 ms in most tests and within about 110 ms in all of them. Stress recorded during this period reflects the instrument's inertia rather than the blood (Das et al., 2024), so it is excluded, with a minimum exclusion of 50 ms.

3. Methods

3.1 Transient features

Each response is reduced to a fixed set of interpretable features, all computed after the motor window. Their definitions are given in Table 1; Figure 3 shows them on two real tests.

Table 1. Transient features computed for every step test.

Feature	Definition	Meaning
Stable value σ_{stable}	Median stress from 30 s to the end of the test	Steady-state stress at the new rate
Overshoot	(highest stress after the step − σ_{stable}) ÷ σ_{stable}	How far stress rises above its final level
Undershoot	(σ_{stable} − lowest stress after the step) ÷ σ_{stable}	How far stress falls below its final level
Initial slope	Slope of σ/σ_{stable} against $\log_{10}$(time), from 0.05 to 1 s	Change per tenfold increase in time early on
Settling time	Time after which stress stays within σ_{stable} ± 10 %	How long the response takes to stabilise
Area	Sum of $(\sigma - \sigma_{stable})/\sigma_{stable}$ over log time	Net excess (+) or deficit (−) of stress
Time of peak / minimum	Time at which the highest / lowest stress occurs	Timing of the extreme

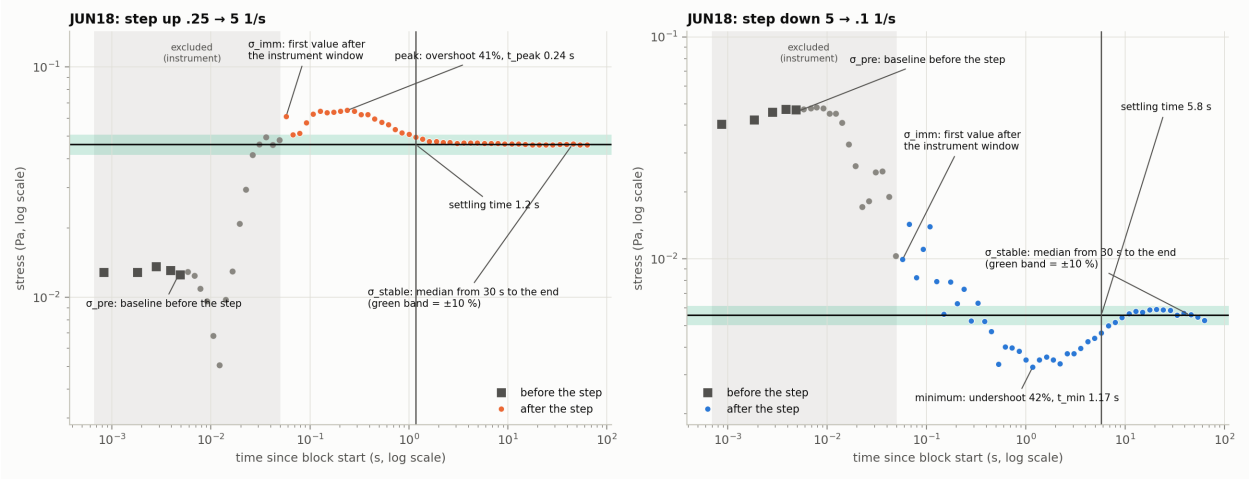


Figure 3. Anatomy of a step response, sample JUN18. Horizontal axis: time since the start of the recording (s, log scale). Vertical axis: stress (Pa, log scale). Squares: before the step; coloured points: after it; grey band: excluded motor window; black line: σ_{stable}; green band: ±10 %. Left: step up 0.25 → 5 s^{-1}. Right: step down 5 → 0.1 s^{-1}.

Worked example. step up 0.25 → 5 s^{-1} (Figure 3, left): σ_{stable} = 0.0461 Pa and the highest stress is 0.0648 Pa at 0.24 s, so overshoot = (0.0648 − 0.0461) ÷ 0.0461 = 0.41 (41 %). The stress enters the ±10 % band for good at 1.2 s.

Worked example. step down 5 → 0.1 s^{-1} (Figure 3, right): σ_{stable} = 0.0056 Pa and the lowest stress is about 0.0033 Pa at 1.17 s, so undershoot ≈ 0.42 (42 %). Settling time is 5.8 s, and the initial slope is −1.30: over each tenfold increase in time during the first second, stress falls by 1.3 times its stable value.

The step-up settles in 1.2 s and the step-down in 5.8 s, roughly five times slower. This asymmetry, fast breakdown and slow rebuilding of rouleaux, is the behaviour described by Pincot et al. (2023).

3.2 Fixed transient and stable windows

The transient and stable portions of each test are labelled by time alone, using a rule fixed before any results were examined: the **transient window** runs from the end of the motor window to 2 s, and the **stable window** from 30 s to the end of the 63 s test. The published structural time of blood is about 1.1 s (Armstrong et al., 2022), so 2 s is roughly two structural times and 30 s more than twenty-five.

Formula. drift = change in stress across the stable window (from a straight-line fit against log time) ÷ σ_{stable}. The stable state is accepted if drift ≤ 10 %; tests that fail are flagged, not removed.

3.3 Thixo-elasto-visco-plastic model

To explain the responses physically, we fitted the scalar thixo-elasto-visco-plastic (ethixo-EVP) model that Armstrong et al. (2021) developed for this type of data. A structure parameter λ between 0 (aggregates fully broken) and 1 (fully structured) is broken down by flow and rebuilt by flow-induced and Brownian aggregation. Writing r for the shear rate:

Formula. steady-state structure: $\lambda_{ss} = (t_{r2}\sqrt{r} + 1) \div (t_{r1}\, r + t_{r2}\sqrt{r} + 1)$

Formula. steady-state stress: $\sigma = \lambda\, \sigma_{y0} + (\lambda\, \eta_{ST} + \eta_\infty)\, r$

Formula. structure kinetics: $d\lambda/dt = k_\lambda [\, -t_{r1}\, r_p\, \lambda + t_{r2}\, \sqrt{r_p}\, (1 - \lambda) + (1 - \lambda)\,]$, with structural time $\tau_\lambda = 1/k_\lambda$

Formula. elastic stress: $d\sigma/dt = G\,(r - r_p)$, where r_p is the plastic (flowing) part of the rate and G the elastic modulus

Following Armstrong et al. (2022), all parameters of a sample were fitted simultaneously to its steady-flow curve and to all of its step tests, minimising squared differences with each step scaled by its own largest stress; the two coupled differential equations were integrated numerically (LSODA, SciPy; Virtanen et al., 2020). The exponent of the aggregation term was fixed at ½, as in all four source papers.

Worked example. for JUN18 the fit gives $t_{r1} = 0.72$, $t_{r2} = 0.13$, $\eta_{ST} = 0.0225$ Pa·s, $\eta_\infty = 0.0032$ Pa·s, $k_\lambda = 1.54\ s^{-1}$ and G = 0.19 Pa. At $r = 5\ s^{-1}$: $\lambda_{ss} = (0.13 \times 2.24 + 1) \div (0.72 \times 5 + 0.13 \times 2.24 + 1) = 1.28 \div 4.89 = 0.26$, and $\sigma = (0.26 \times 0.0225 + 0.0032) \times 5 = 0.046$ Pa, which matches the measured stable stress of 0.0461 Pa. At $r = 0.1\ s^{-1}$, $\lambda_{ss} = 0.94$: slowing from 5 to 0.1 s^{-1} lets the structure rebuild from 0.26 to 0.94.

Formula. fit quality per step: NRMSE = √(mean of (data − model)²) ÷ largest stress of that step; a step is well fitted if NRMSE ≤ 0.10.

As a simpler comparison, a single exponential $\sigma(t) = \sigma_\infty + A \exp(-t/\tau)$ was fitted to each step, and rejected if τ reached a bound, its 95 % interval was wider than τ itself, or $R^2 < 0.5$.

3.4 Statistical analysis

Because many tests come from the same blood, the blood sample is the independent unit throughout; tests and windows are never counted as independent observations.

- **Step magnitude.** Within each family, step size is $|\log_{10}(\text{final rate} \div \text{starting rate})|$ (1 = tenfold change), and the Spearman rank correlation ρ between step size and each feature is computed. Correlations are averaged per sample and a sign test asks whether their sign is consistent across samples.
- **Step direction.** Only the 22 matched pairs are used; for each pair we record which member has the larger feature value, average per sample, and apply a sign test.
- **Transient vs stable.** Three descriptors are computed in each window of each test, compared within the test, and summarised per sample.
- **Direction rules.** Accuracy is reported with a 95 % confidence interval from a bootstrap that resamples whole blood samples 2,000 times.

Formula. Spearman $\rho = 1 - 6\,\Sigma d^2 \div (n(n^2 - 1))$, where d is the difference between the two ranks of each item.

Formula. sign test: if k of n samples show the same sign, $p = 2 \times (½)^n$ when k = n.

4. Results

4.1 Transient and stable phases are distinct

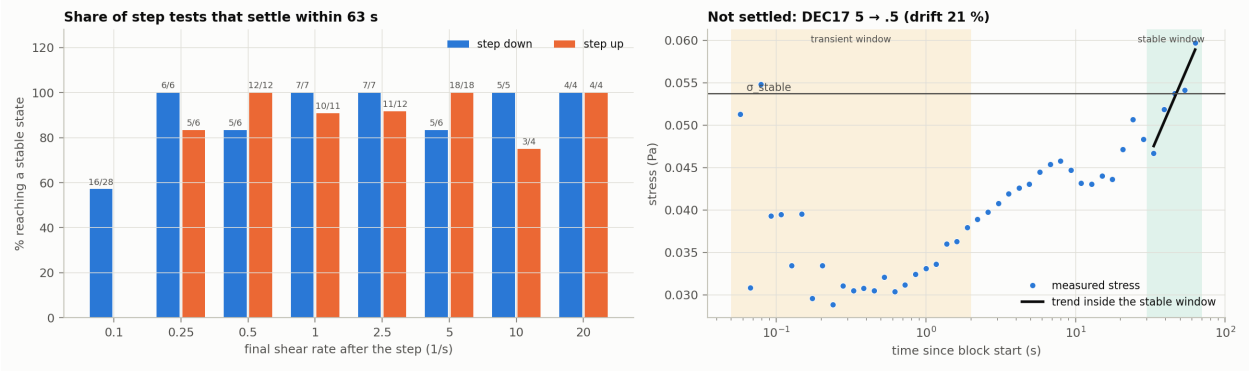


Figure 4. Left: share of tests reaching a stable state (vertical, %) by final shear rate (horizontal, s^{-1}); blue: step-downs, orange: step-ups; counts above bars. Right: a test that does not settle, DEC17 after 24 h of storage, 5 → 0.5 s^{-1}; horizontal axis time (s, log scale), vertical axis stress (Pa); yellow: transient window, green: stable window; black line: trend inside the stable window.

Overall, 118 of 136 tests (87 %) reach a defensible stable state (Figure 4, left). Most groups lie between 75 % and 100 %; the exception is steps down to 0.1 s^{-1}, which settle in only 16 of 28 tests, so their 'stable' values are lower bounds. In the example on the right the stress drops to about 0.03 Pa within half a second, climbs to about 0.045 Pa by 10 s and is still rising at the end; its drift is 21 %, above the 10 % limit.

Three descriptors were computed in each window: the magnitude of the slope of σ/σ_{stable} against log time, the mean absolute deviation $|\sigma - \sigma_{stable}|/\sigma_{stable}$, and the coefficient of variation. All three are larger in the transient window than in the stable window in **all 11 samples** (sign test $p = 2 \times (½)^{11} \approx 0.001$). Repeating the analysis with shorter (1 s / 20 s) and longer (5 s / 40 s) windows gives the same result, except that the slope descriptor drops to 9 of 11 samples with the longest transient window, because most responses have flattened by 5 s.

4.2 The model reproduces published parameters

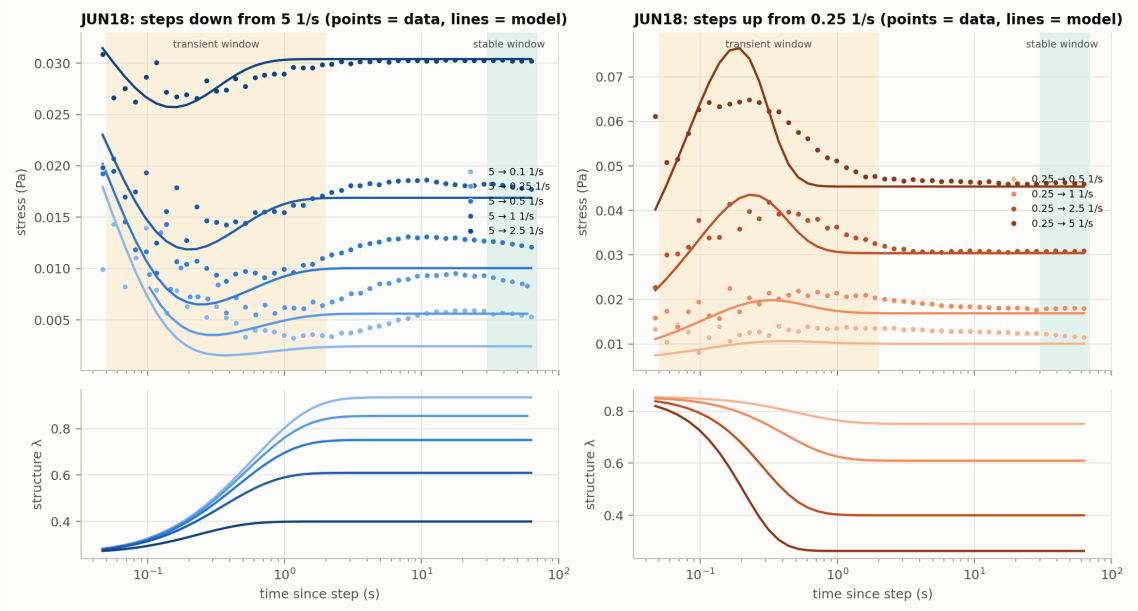


Figure 5. Model fits for JUN18. Horizontal axis: time since the step (s, log scale). Top: stress (Pa); points are data, lines the model, darker shades larger final rates; left steps down from 5 s^{-1}, right steps up from 0.25 s^{-1}. Bottom: the model's structure parameter λ.

The model reproduces the dip-and-recovery of step-downs and the overshoot of step-ups (Figure 5). After a step down, λ rises from about 0.28 to between 0.40 and 0.94 depending on the final rate (rebuilding); after a step up it falls from about 0.85 to between 0.75 and 0.26 (breakdown), faster for larger steps. All 13 sample-level fits (11 samples, DEC17 at three storage times) converged with no parameter at a bound. Table 2 compares the fitted parameters with the literature.

Table 2. Fitted model parameters compared with the literature.

Parameter	This analysis	Published values
Structural time τλ	median 0.64 s (IQR 0.27–1.16 s); 5 of 13 fits within the 12-donor mean ± 2 SD	1.13 ± 0.24 s over 12 donors (Armstrong et al., 2022); 0.55–0.96 s for one donor (Pincot & Armstrong, 2021)
Elastic modulus G	median 0.27 Pa (range 0.19–0.47 Pa)	0.29 Pa for the same model (Armstrong et al., 2021)

4.3 Where the model falls short

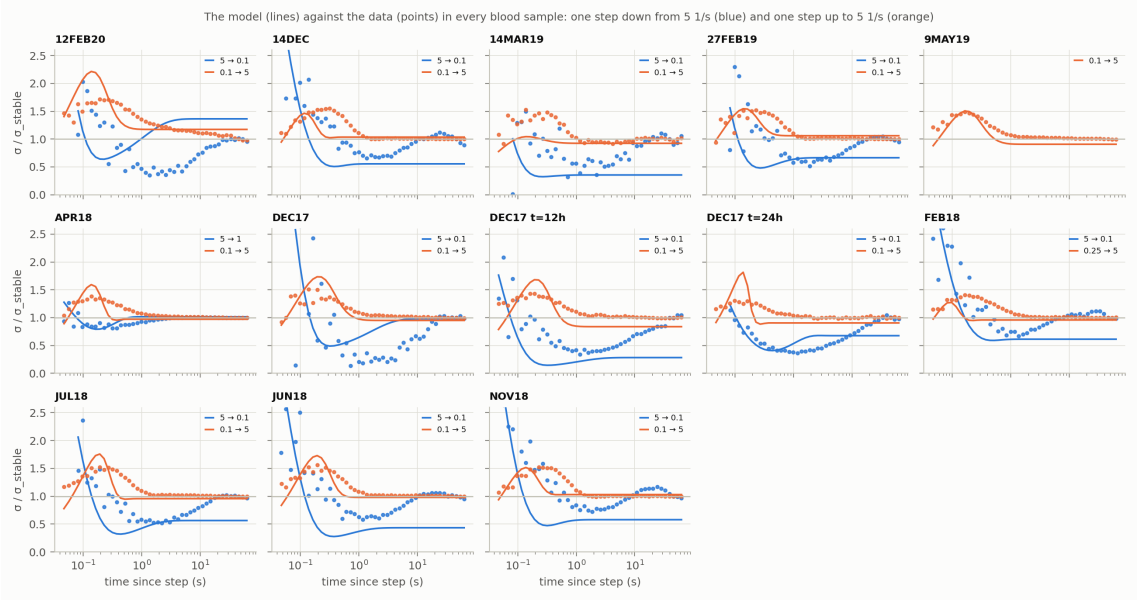


Figure 6. Model against data in every sample. Horizontal axis: time since the step (s, log scale). Vertical axis: stress divided by the step's stable value (dimensionless; 1 = steady state). Blue: a step down from 5 s^{-1}; orange: a step up to 5 s^{-1}. Points are data, lines the model.

For step-ups the model peaks at about the right time (Figure 6, orange). For step-downs it does not capture the full recovery: the data fall to about half the stable value within a second and then climb back to 1 over 10–60 s, whereas the model levels off within about a second at 0.3–0.7 of the stable value. In JUN18, for example, the model flattens at about 0.4 while the data reach 1 at about 10 s. Only 50 of 133 steps meet the NRMSE ≤ 0.10 criterion, with the poorest fits on steps ending at low rates (median NRMSE 0.17–0.21 at 0.1–0.25 s^{-1}, against 0.05–0.08 for final rates of 10 s^{-1} or more). The single exponential is worse as a metric: only 49 % of steps give an interpretable τ, because a response that dips and then recovers cannot be described by one exponential.

This points to a second, slower rebuilding process that a model with a single structure parameter cannot represent, consistent with multi-time-scale thixotropy models (Wei et al., 2018). Model outputs are therefore used only as one value per sample, not interpreted step by step.

4.4 Step size and direction

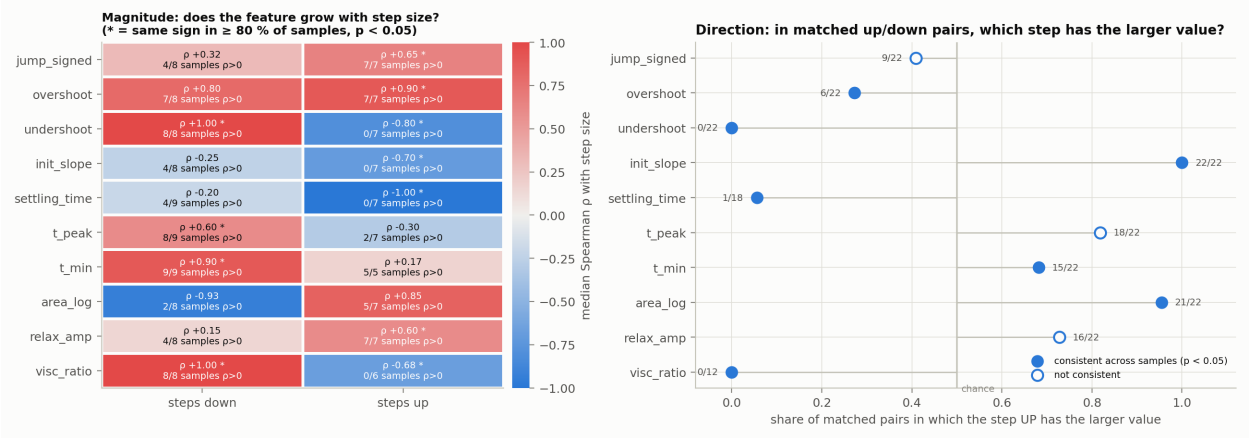


Figure 7. Evidence summary. Left: median Spearman ρ between each feature (rows) and step size, for steps down and steps up (columns); red = rises with step size, blue = falls; * = same sign in ≥ 80 % of samples with p < 0.05. Right: share of the 22 matched pairs (horizontal, 0–1) in which the step up has the larger value; 0.5 = chance; filled dots are consistent across samples.

> **Worked example.** JUN18, steps down from 5 s^{-1} to 2.5, 1, 0.5, 0.25 and 0.1 s^{-1} have step sizes 0.30, 0.70, 1.00, 1.30 and 1.70 and undershoots 0.13, 0.30, 0.29, 0.37 and 0.42. Ranked, the undershoots go 1, 3, 2, 4, 5 against step-size ranks 1–5, so $\Sigma d^2 = 2$ and $\rho = 1 - 6 \times 2 \div (5 \times 24) = 0.9$. Across the 8 samples with step-down families every ρ is positive, giving $p = 2 \times (½)^8 = 0.008$.

Magnitude (Figure 7, left) is encoded by different features in each direction. For step-downs, undershoot, the time of the minimum and the viscosity ratio rise with step size in every sample (8/8, 9/9 and 8/8). For step-ups, overshoot, the signed jump and the relaxation amplitude rise, and the initial slope, settling time and undershoot fall, in all 7 samples. The area feature is not consistent in either direction.

> **Worked example.** JUN18, 5 → 0.1 against 0.1 → 5 s^{-1}: undershoot 0.42 for the step down against 0.005 for the step up; initial slope −1.30 against −0.10. The step down has the larger undershoot and the steeper initial fall.

Direction (Figure 7, right): in the matched pairs, the step down has the larger undershoot in 22 of 22 pairs (10 of 10 samples, p = 0.002), the steeper initial fall in 22 of 22, and the step up the larger area in 21 of 22. The signed jump, the feature originally expected to carry direction, is near chance (9 of 22).

4.5 Rules for telling step up from step down

Formula. pre-specified rule: signed jump = (first stress after the motor window − σ_{stable}) ÷ σ_{stable}; classify the step as 'up' if the signed jump is above 0, otherwise 'down'.

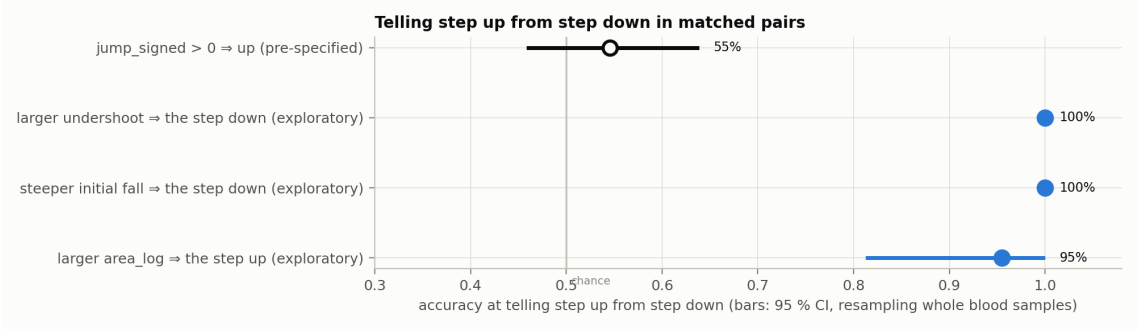


Figure 8. Accuracy of four direction rules on the matched pairs (horizontal, 0.3–1.0); bars are 95 % confidence intervals from resampling whole blood samples; 0.5 = chance. Hollow black: rule fixed in advance; blue: rules found after inspecting the data.

The pre-specified rule achieves 55 % accuracy (95 % CI 46–64 %), no better than chance (Figure 8).

> **Worked example.** JUN18, step down 5 → 0.1 s^{-1}: the first stress after the motor window is 0.0099 Pa and σ_{stable} = 0.0056 Pa, so the signed jump is (0.0099 − 0.0056) ÷ 0.0056 = +0.77 and the rule calls the step 'up'. Just after a step down, the stress is still relaxing elastically from its earlier, higher level, so the first value is misleading.

Three rules identified after inspecting the data perform far better: 'the step with the larger undershoot is the step down' and 'the step with the steeper initial fall is the step down' are correct in all 22 pairs, and 'the step with the larger area is the step up' in 21 of 22 (95 %, CI about 81–100 %). Because these rules were chosen on the same data, they are hypotheses that require confirmation on new samples, not validated classifiers.

4.6 Which metrics can be trusted

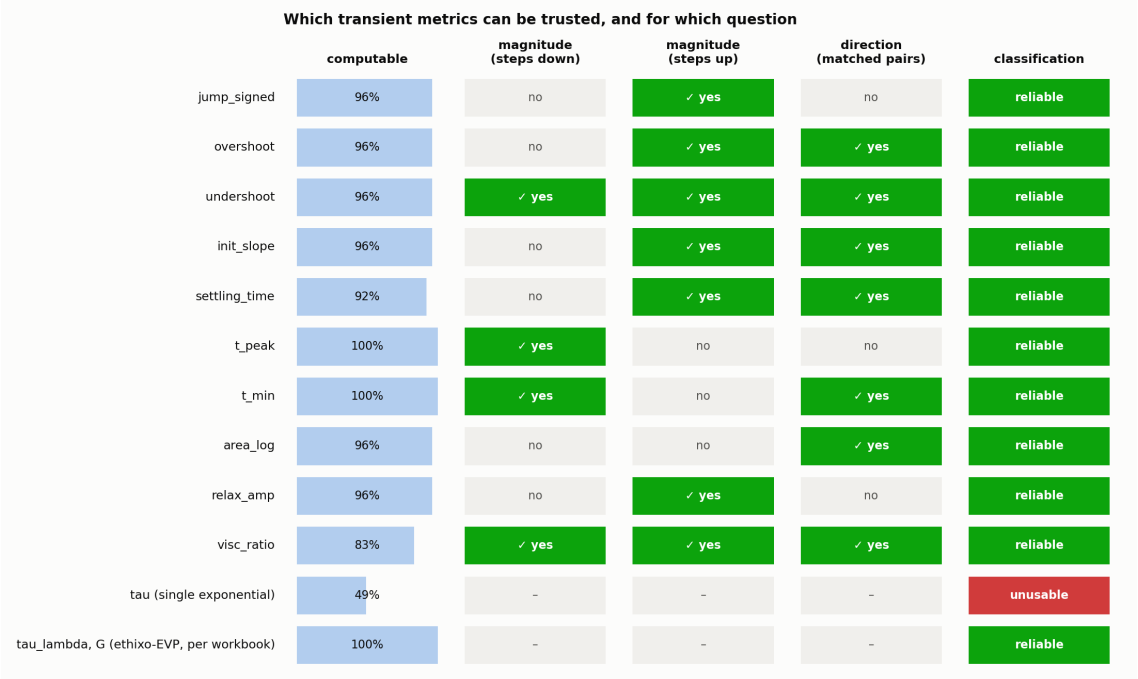


Figure 9. Metric reliability. Rows: metrics. Columns: share of steps where the metric is computable; whether it answers each question (tick: same sign in ≥ 80 % of samples and $p < 0.05$); overall classification.

A metric is classed reliable if it is computable for at least 80 % of steps and answers at least one question, and unusable if it is computable for fewer than 50 % (Figure {n}). Undershoot and the viscosity ratio answer all three questions; overshoot, initial slope and settling time answer step-up magnitude and

direction; the times of the peak and minimum mainly answer step-down magnitude. The single-exponential τ is computable for only 49 % of steps and is unusable. The model parameters τ_λ and G are reliable, but only as one value per sample.

5. Discussion and recommendations

Candidate real-time features. Undershoot, initial slope and settling time carry most of the information about step size and direction and need only the first seconds of a step and very little computation. On a new step they can be obtained as follows:

- record at least 30 s after the step, plus a few points before it;
- discard the first 50 ms after the step;
- σ_{stable} = median stress over the last 30 s;
- undershoot = (σ_{stable} − minimum stress) ÷ σ_{stable};
- initial slope = slope of σ/σ_{stable} against $\log_{10}$(time) between 0.05 and 1 s;
- settling time = last time the stress lies outside σ_{stable} ± 10 %.

  Worked example. for JUN18, step down 5 → 0.1 s^{-1}, these give an undershoot of 0.42, an initial slope of −1.30 per decade and a settling time of 5.8 s.

The viscosity ratio also scores well but partly restates the step itself, because blood's viscosity rises when the rate falls, so it is a weaker candidate.

Offline characterisation. τ_λ and G agree with the literature and describe how fast and how stiff the structure is, but they require a flow curve and a model fit, so they suit offline characterisation rather than real-time use.

Methodological lessons. Fixing the analysis rules in advance exposed a plausible direction rule that does not work; grouping all statistics by blood sample prevented inflated evidence from many correlated tests; and reporting the failed rule alongside the exploratory successes keeps the conclusions honest.

Data quality. The verification step found one mislabelled block (APR18 '5 → 0.1', actually 5 → 0.5), one test that did not start at its stated rate (14MAR19 '0.1 → 10'), and copied blocks inside several workbooks, including one where only the stress column was copied (DEC17 24 h '0.1 → 1'). Checking both the starting and the final rate, and comparing stress traces on their own, is recommended for any shared processing pipeline.

Limitations. There are only 11 independent blood samples, so all results are exploratory. Steps down to 0.1 s^{-1} often do not settle within 63 s, so their stable values are lower bounds. The model uses a single structure parameter and cannot reproduce the slow recovery after step-downs. Most tests have only 65 log-spaced points, which limits the time resolution available to signal-transform methods.

Next steps. The exploratory direction rules should be tested on new blood samples; Fourier and wavelet features should be computed on exactly the same steps and windows so that they can be compared fairly with the simple features reported here; and a model with more than one structural time scale could be tried to capture the slow recovery.

6. Conclusion

Step tests on human blood carry clear, interpretable information. The transient and stable phases differ in every sample; step size is encoded by undershoot and minimum time for step-downs and by overshoot and settling time for step-ups; and the undershoot separates step-down from step-up in every matched pair. A published thixo-elasto-visco-plastic model reproduces the literature values of the structural time and stiffness on these data but misses a slower recovery process. Three cheap features, undershoot, initial slope and settling time, are recommended as candidates for real-time use, subject to confirmation on new samples.

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Five of the workbooks analysed (12FEB20, 27FEB19, NOV18, 14DEC, AUG18) have no published DOI in the source papers. The analysis code and all figures are in the accompanying notebook *Step_UpDown_Analysis.ipynb*.